

Objectives lab18 11.5.1 Packet Tracer - Compare Layer 2 and Layer 3 Devices - ILM

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Part 1: Compare Layer 2 and Layer 3 Switches

Part 2: Compare a Layer 3 Switch and a Router

Background / Scenario

In this activity, you will use various commands to examine three different switching topologies and compare the similarities and differences between the 2960 and 3650 switches. You will also compare the routing table of a 4321 router with that of a 3650 switch.

Note: Search the internet for more details about the *WS-C3650-24PS-L Layer 3 switch* and the *ISR 4321/K9 router*.

Instructions

Part 1: Compare Layer 2 and Layer 3 Switches

Step 1: Explore the Physical Workspace.

- In the Physical Workspace, click the **Home City** image. Click the **Corporate Office** image. Click the **Main Wiring Closet** image.
- In the Rack, locate the devices **D1** and **ASw-1**. Examine the physical aspects of **D1** and **ASw-1**. If you need to examine the devices more closely, click the device and select the Physical tab.

Each individual switch has how many physical switchports?

Device Name: D1
Device Model: 3650-24PS
Hostname: D1

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet1/0/1	Down	1	<not set>	<not set>	0005.5E00.2801
GigabitEthernet1/0/2	Down	1	<not set>	<not set>	0005.5E00.2802
GigabitEthernet1/0/3	Down	1	<not set>	<not set>	0005.5E00.2803
GigabitEthernet1/0/4	Down	1	<not set>	<not set>	0005.5E00.2804
GigabitEthernet1/0/5	Down	1	<not set>	<not set>	0005.5E00.2805
GigabitEthernet1/0/6	Down	1	<not set>	<not set>	0005.5E00.2806
GigabitEthernet1/0/7	Down	1	<not set>	<not set>	0005.5E00.2807
GigabitEthernet1/0/8	Down	1	<not set>	<not set>	0005.5E00.2808
GigabitEthernet1/0/9	Down	1	<not set>	<not set>	0005.5E00.2809
GigabitEthernet1/0/10	Down	1	<not set>	<not set>	0005.5E00.280A
GigabitEthernet1/0/11	Down	1	<not set>	<not set>	0005.5E00.280B
GigabitEthernet1/0/12	Down	1	<not set>	<not set>	0005.5E00.280C
GigabitEthernet1/0/13	Down	1	<not set>	<not set>	0005.5E00.280D
GigabitEthernet1/0/14	Down	1	<not set>	<not set>	0005.5E00.280E
GigabitEthernet1/0/15	Down	1	<not set>	<not set>	0005.5E00.280F
GigabitEthernet1/0/16	Down	1	<not set>	<not set>	0005.5E00.2810
GigabitEthernet1/0/17	Down	1	<not set>	<not set>	0005.5E00.2811
GigabitEthernet1/0/18	Down	1	<not set>	<not set>	0005.5E00.2812
GigabitEthernet1/0/19	Down	1	<not set>	<not set>	0005.5E00.2813
GigabitEthernet1/0/20	Down	1	<not set>	<not set>	0005.5E00.2814
GigabitEthernet1/0/21	Down	1	<not set>	<not set>	0005.5E00.2815
GigabitEthernet1/0/22	Down	1	<not set>	<not set>	0005.5E00.2816
GigabitEthernet1/0/23	Down	1	<not set>	<not set>	0005.5E00.2817
GigabitEthernet1/0/24	Down	1	<not set>	<not set>	0005.5E00.2818
GigabitEthernet1/1/1	Up	1	192.168.10.1/24	<not set>	0030.F2DB.5D01
GigabitEthernet1/1/2	Up	1	192.168.11.1/24	<not set>	0030.F2DB.5D02
GigabitEthernet1/1/3	Down	1	<not set>	<not set>	0030.F2DB.5D03
GigabitEthernet1/1/4	Down	1	<not set>	<not set>	0030.F2DB.5D04
Vlan1	Down	1	<not set>	<not set>	0005.5EB3.7A2E

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > D1

Device Name: ASw-1				
Device Model: 2960-24TT				
Hostname: ASw-1				
Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	1	--	0040.0B7D.B901
FastEthernet0/2	Down	1	--	0040.0B7D.B902
FastEthernet0/3	Down	1	--	0040.0B7D.B903
FastEthernet0/4	Down	1	--	0040.0B7D.B904
FastEthernet0/5	Down	1	--	0040.0B7D.B905
FastEthernet0/6	Down	1	--	0040.0B7D.B906
FastEthernet0/7	Down	1	--	0040.0B7D.B907
FastEthernet0/8	Down	1	--	0040.0B7D.B908
FastEthernet0/9	Down	1	--	0040.0B7D.B909
FastEthernet0/10	Down	1	--	0040.0B7D.B90A
FastEthernet0/11	Down	1	--	0040.0B7D.B90B
FastEthernet0/12	Down	1	--	0040.0B7D.B90C
FastEthernet0/13	Down	1	--	0040.0B7D.B90D
FastEthernet0/14	Down	1	--	0040.0B7D.B90E
FastEthernet0/15	Down	1	--	0040.0B7D.B90F
FastEthernet0/16	Down	1	--	0040.0B7D.B910
FastEthernet0/17	Down	1	--	0040.0B7D.B911
FastEthernet0/18	Down	1	--	0040.0B7D.B912
FastEthernet0/19	Down	1	--	0040.0B7D.B913
FastEthernet0/20	Down	1	--	0040.0B7D.B914
FastEthernet0/21	Down	1	--	0040.0B7D.B915
FastEthernet0/22	Down	1	--	0040.0B7D.B916
FastEthernet0/23	Down	1	--	0040.0B7D.B917
FastEthernet0/24	Down	1	--	0040.0B7D.B918
GigabitEthernet0/1	Up	1	--	0040.0B7D.B919
GigabitEthernet0/2	Down	1	--	0040.0B7D.B91A
Vlan1	Up	1	192.168.10.2/24	0001.C92E.00C5
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > ASw-1				

2960: มี 26 พอร์ต

3650: มี 24 พอร์ตแบบตายตัว (fixed) + 4 พอร์ตแบบโมดูลาร์ที่รองรับ

SFP moduleHow many Fast Ethernet and Gigabit Ethernet switchports does each switch have?

2960: Fast Ethernet 24 พอร์ต + Gigabit Ethernet 2 พอร์ต

3650: Gigabit Ethernet ล้วน — 24 พอร์ตตายตัว + 4 พอร์ตโมดูลาร์

List the transmission speed of the Fast Ethernet and Gigabit Ethernet switchports on each switch.

Fast Ethernet: 10/100 Mbps

Gigabit Ethernet: สูงสุด 1000 Mbps

Are either of the two switches modular in design?

3650 เท่านั้นที่เป็นโมดูลาร์ (เพราะมีสล็อต SFP ให้เสียบโมดูลเพิ่ม)

Step 2: In the CLI tab

- a. The switchports of a 3650 switch can be configured as Layer 3 interfaces by entering the **no switchport** command in interface configuration mode. This allows technicians to assign an IP address and subnet mask to the switchport in the same way that they are configured on a router interface.

What is the difference between a Layer 2 switch and a Layer 3 switch?

L2 switch ตัดสินใจส่งข้อมูล (forwarding) โดยใช้ MAC address เท่านั้น

L3 switch สามารถกำหนด IP address ให้กับพอร์ตทางกายภาพได้ (ผ่านคำสั่ง no switchport) และรองรับ routing protocol ได้เหมือน router

What is the difference between a switch's physical interface and the VLAN interface?

Physical interface: ใช้เชื่อมต่ออุปกรณ์ปลายทางเข้าเครือข่ายจริง

VLAN interface (SVI): เป็น interface เสมือนที่ใช้กำหนด IP ให้สวิตช์ เพื่อบริหารจัดการ (manage) จาก ระยะไกล

At which layers do 2960 and 3650 switches operate?

2960: Layer 2 เท่านั้น

3650: Layer 2 และ Layer 3

- a. Navigate to the CLI tab for both devices. Issue the **show run** command to examine the configurations of the **D1** and **ASw-1** switches.

Do you notice any differences between them?

D1 ใช้ Gigabit Ethernet ทุกพอร์ต ส่วน 2960 ส่วนใหญ่เป็น Fast Ethernet และมี Gigabit 2 พอร์ตไว้ใช้ uplink

D1 ใช้รูปแบบการตั้งชื่อพอร์ตแบบ stack-module-port (เช่น G1/1/1)

D1 มีพอร์ตที่ตั้งค่า no switchport และกำหนด IP address ไว้บน G1/1/1, G1/1/2

D1 เปิดใช้ ip routing ในขณะที่ ASw-1 ไม่มี

- a. Try to display the routing table on D1 and ASw-1 using the **show ip route** command.

Why do you think the command does not work on **ASw-1** but works on **D1**?

```

D1>ena
D1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet1/1/1
L       192.168.10.1/32 is directly connected, GigabitEthernet1/1/1
    192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet1/1/2
L       192.168.11.1/32 is directly connected, GigabitEthernet1/1/2

D1#

```

```

ASw-1>
ASw-1>ena
ASw-1#show ip route
      ^
% Invalid input detected at '^' marker.

ASw-1#

```

D1 ทำงานทั้ง Layer 2 และ Layer 3 จึงมี routing table

ASw-1 เป็น L2 switch ธรรมดาๆ จึงไม่มี routing table เลย

Part 2: Compare a Layer 3 Switch and a Router

In the past, switches and routers have been separate and distinct devices. The term switch was set aside for hardware devices that function at Layer 2. Routers, on the other hand, are devices that make forwarding decisions based on Layer 3 information. They use routing protocols to share routing information and to communicate with other routers. Layer 3 switches, such as the 3650, can be configured to forward Layer 3 packets. Entering the **ip routing** command in global configuration mode allows Layer 3 switches to be configured with routing protocols, which gives them some of the capabilities of a router. Although similar in some ways, Layer 3 switches are different from routers in many other aspects.

Step 1: Compare D1 and R1

- Open the Physical tab on D1 and R1.

Do you notice any similarities between the two? Do you notice any differences between the two?

เหมือนกัน: มี console port, USB port, Gigabit Ethernet interface, และเป็นแบบโมดูลาร์ทั้งคู่

ต่างกัน: R1 มี Serial และ Asynchronous interface ด้วย ส่วน D1 มีแต่ Ethernet (ทองแดงหรือไฟเบอร์ ขึ้นกับโมดูล) และ D1 มีจำนวนพอร์ต Gigabit มากกว่า R1 มาก

- b. In the CLI tab, issue the **show run** command and examine the configurations of R1 and D1.

What differences do you see between the two?

ทั้งคู่ตั้งค่า IP address เดียวกัน แต่คนละ interface

Which command allows configuration of D1 with an IP address on one of its physical interfaces?

คำสั่ง no switchport

- c. Use the **show ip route** command on both devices.

Do you see any similarities or differences between the two tables?

```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
    192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.11.1/32 is directly connected, GigabitEthernet0/0/1

R1#
```

โค้ดในตารางเหมือนกันเกือบทั้งหมด ยกเว้น router มีโค้ด L (local) ซึ่งเป็นลิงก์ที่ตั้งค่าบน physical interface ของ R1 โดยตรง ส่วนสวิตช์ไม่มีโค้ดนี้

ทั้งสองอุปกรณ์เห็นเครือข่ายเดียวกันในตาราง routing Compare R2 and D2

- d. Now, analyze the routing table of R2 and D2. Click both devices in the Rack. In the CLI tab, issue the **show ip route** command.

What is present now that was not present in the configuration of R1 and D1?

```

R2>
R2>ena
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       1.1.1.0/24 is directly connected, GigabitEthernet0/0/1
L       1.1.1.1/32 is directly connected, GigabitEthernet0/0/1
    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.11.48.0/30 is directly connected, GigabitEthernet0/0/0
L       10.11.48.1/32 is directly connected, GigabitEthernet0/0/0
O       192.168.10.0/24 [110/2] via 10.11.48.2, 01:32:59, GigabitEthernet0/0/0
O       192.168.11.0/24 [110/2] via 10.11.48.2, 01:32:59, GigabitEthernet0/0/0

R2#

```

```

D2>
D2>ena
D2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    1.0.0.0/24 is subnetted, 1 subnets
O       1.1.1.0 [110/2] via 10.11.48.1, 01:33:38, GigabitEthernet1/0/1
    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.11.48.0/30 is directly connected, GigabitEthernet1/0/1
L       10.11.48.2/32 is directly connected, GigabitEthernet1/0/1
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet1/1/1
L       192.168.10.1/32 is directly connected, GigabitEthernet1/1/1
    192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet1/1/2
L       192.168.11.1/32 is directly connected, GigabitEthernet1/1/2

D2#

```

ทั้งคู่ว่าน OSPF และเรียนรู้เส้นทาง (route) จากกันและกัน

Which network is in the routing table of D2 that was learned from R2?

เครือข่าย 1.1.1.0/24

- e. Click the **Back Level** icon (Alt + Left) to leave the Main Wiring Closet. Verify that each topology has full connectivity by completing the following tests:

Ping from **PC1** to **PC2**

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.10

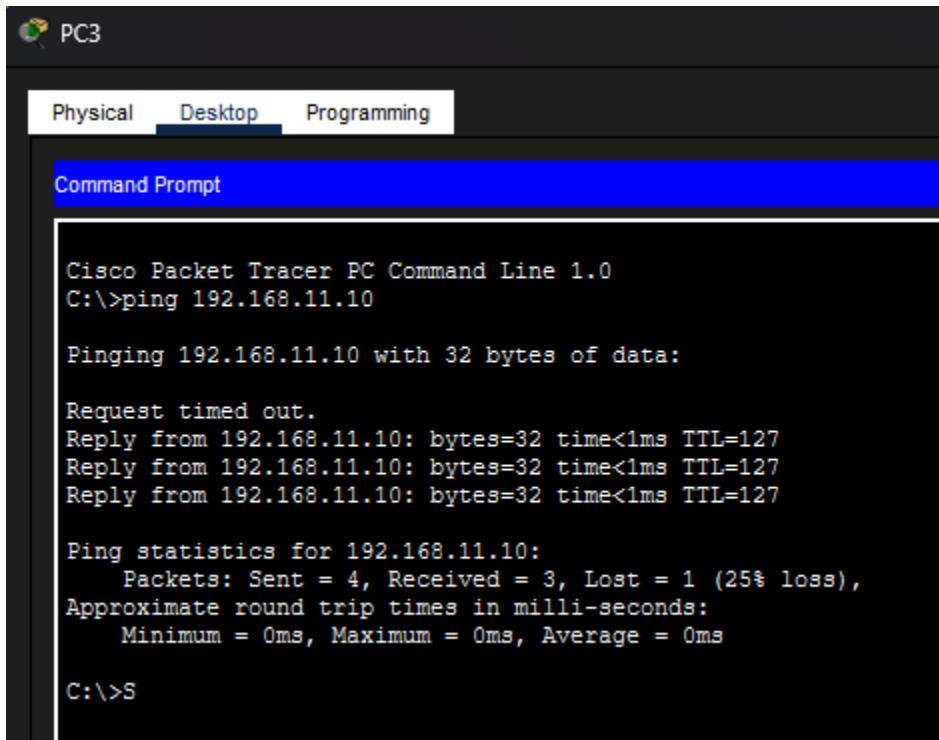
Pinging 192.168.11.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.11.10: bytes=32 time<1ms TTL=127
Reply from 192.168.11.10: bytes=32 time<1ms TTL=127
Reply from 192.168.11.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.11.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Ping from **PC3** to **PC4**



The screenshot shows the 'PC3' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The text in the Command Prompt is identical to the first screenshot, showing a ping from PC3 to PC4 (192.168.11.10) with a 25% loss. The prompt ends with 'C:\>S'.

```
PC3

Physical Desktop Programming

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.10

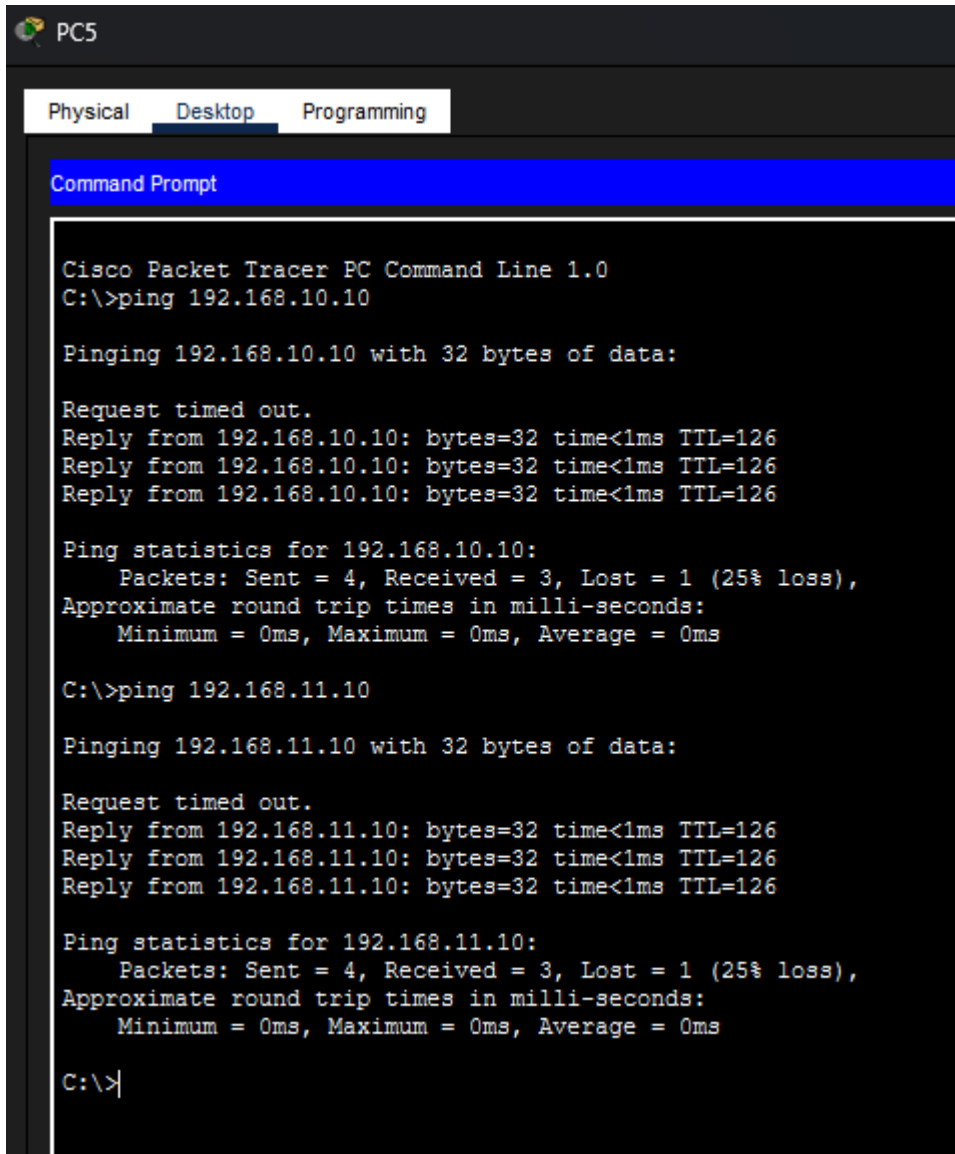
Pinging 192.168.11.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.11.10: bytes=32 time<1ms TTL=127
Reply from 192.168.11.10: bytes=32 time<1ms TTL=127
Reply from 192.168.11.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.11.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>S
```

Ping from **PC5** to **PC6** and **PC7**



```
PC5
Physical Desktop Programming
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.11.10

Pinging 192.168.11.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.11.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

In all three examples, each PC is on a different network.

Note: You may need to move the PCs to see them.

Which device is used to provide communication between networks?

Router หรือ multilayer switch

Why were we able to ping across networks without there being a router?

Multilayer switch สามารถ route ข้ามเครือข่ายได้ トラバไตที่ตั้งค่า IP address และเปิด ip routing ไว้ และถ้าจะรัน routing protocol อย่าง OSPF ก็ต้องเปิด ip routing เช่นกัน ส่วนพอร์ตที่จะกำหนด IP ได้ต้องใช้คำสั่ง no switchport ก่อน

Bonus question: We say that routers are Layer 3 devices and conventional (non-Layer 3) switches are Layer 2 devices. However, we can assign an IP address to a management (SVI) interface of a Layer 2 switch. How is this possible if switches are Layer 2 devices?

สวิตช์ L2 อย่าง Catalyst 2960 มี embedded server ในตัว ที่เข้าถึงได้ผ่าน Layer 3 (Telnet/SSH/HTTP) เพื่อใช้บริหารจัดการระยะไกล ฟังก์ชันนี้แตกต่างจากฟังก์ชัน forwarding ข้อมูลที่ยังอยู่ที่ Layer 2 เหมือนเดิม

ทดสอบ **show interfaces status** บน D1 vs ASw-1

D1

```
D1>
D1>ena
D1#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gig1/0/1		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/2		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/3		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/4		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/5		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/6		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/7		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/8		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/9		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/10		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/11		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/12		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/13		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/14		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/15		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/16		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/17		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/18		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/19		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/20		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/21		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/22		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/23		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/24		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/1/1		connected	routed	a-full	a-100	10/100/1000BaseTX
Gig1/1/2		connected	routed	a-full	a-100	10/100/1000BaseTX
Gig1/1/3		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/1/4		notconnect	1	auto	auto	10/100/1000BaseTX

```
D1#
```

ASw-1

```

ASw-1>
ASw-1>ena
ASw-1#show interfaces status

```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	1	a-full	a-100	10/100BaseTX
Fa0/2		notconnect	1	auto	auto	10/100BaseTX
Fa0/3		notconnect	1	auto	auto	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		notconnect	1	auto	auto	10/100BaseTX
Fa0/24		notconnect	1	auto	auto	10/100BaseTX
Gig0/1		connected	1	a-full	a-100	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

```

ASw-1#

```

ทดลอง show vlan brief

```

ASw-1>
ASw-1>ena
ASw-1#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

ASw-1#

```

```

D1>ena
D1#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Gig1/0/1, Gig1/0/2, Gig1/0/3, Gig1/0/4 Gig1/0/5, Gig1/0/6, Gig1/0/7, Gig1/0/8 Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12 Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16 Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20 Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24 Gig1/1/3, Gig1/1/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

D1#

```

ประเด็นเชิงสถาปัตยกรรม

Layer 3 switch เหมาะสำหรับงาน routing ความเร็วสูงภายใน LAN/Campus (inter-VLAN routing) เพราะทำ hardware-based routing (ASIC) แต่มักไม่มี WAN interface (Serial, DSL ฯลฯ) เหมือน router — router ยังจำเป็นสำหรับการเชื่อมต่อ WAN/Internet หรือ feature ที่ซับซ้อนกว่า เช่น NAT ขั้นสูง, VPN, QoS ระดับ WAN